

Adiraju Aditya Srivatsa

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Machine Learning & Robotics Engineer | Passionate about applying areas related to AI automation, data-driven intelligence to real-world electromechanical systems, bridging ML and electrical systems, with hands-on experience in ROS2

Education

Technische Hochschule Würzburg-Schweinfurt – B.Eng in Robotics

Graduation: March 2027

- **Notable Coursework:** Machine Learning, Statistics, Control Systems, Image Processing, Electronics, Embedded Systems, Software Engineering
- **Relevant Labs:** Line Follower, Control Systems, MATLAB & Simulink, ROS, Camera Calibration

Experience

Intern - OTIS Lead Design Center Berlin

Intern, E&E Engineering - August 2025 - January 2026

- Led development of a machine-learning pipeline for CAN-bus trace analysis, automating root-cause detection for elevator faults, a first-of-its-kind approach within the team.
- Achieved 96 % diagnostic accuracy and reduced manual fault analysis time.
- Integrated SHAP-based explainable AI to make model insights transparent and actionable for cross-disciplinary teams.
- Rewrote a Shell-based firmware automation script in Python (HEX / LD file generation), improving build reliability and simplifying maintenance.
- Independently led the full project lifecycle, including planning, testing and development in an agile-inspired workflow.
- Presented results and a proposal for integration into OTIS's predictive-maintenance platform.

Projects

1. Root Cause Detection via CAN trace Analysis - OTIS Internship

- Designed and deployed a machine learning solution using HistGradientBoosting and SHAP to identify root causes from complex CAN trace data. Leveraged GenAI tools to accelerate code development and testing. Designed to reduce manual analysis time. Delivered the full project independently and proposed scalable extensions.

2. Stock Market Outperformance Prediction & ML-Based Investment Analysis

GitHub Code Repository: [Stock_Analysis](#)

- Developed an end-to-end machine learning pipeline to predict stock outperformance using financial fundamentals, technical indicators, SEC filing sentiment, and market time-series data. Engineered features from quarterly reports, OHLCV data, and 8-K filings using NLP techniques (VADER, TF-IDF, TruncatedSVD), stored and managed datasets with SQL, and trained optimized XGBoost and Random Forest models using time-series cross-validation and backtesting. Built a web dashboard to visualize predictions, model performance, feature importance, and investment insights.

3. Collision Detection System

GitHub Code Repository: [ML_Automotive_Safety](#)

- Developed collision detection using YOLOv8 and sensor fusion; retrofitted older vehicles with low-cost safety alerts involving hands-on hardware assembly and sensor integration. Presented at Campus Startup Night by Werkraum, THWS.

4. Warehouse Tracking System

GitHub Code Repository: [Multi_Camera_Tracking](#)

- Developed a multi-camera person-tracking system for enhancing warehouse safety using YOLOv8, DeepSORT, and homography transformation to map real-world positions across CCTV feeds. Achieved reliable multi-person ID tracking and top-down spatial visualization for monitoring worker movement and improving safety awareness. Developed collaboratively with cross-functional input and submitted as a prototype to **Sentics GmbH**.

Skills

Programming & Tooling: Python, C++, ROS2, C, Java, Shell Scripting, SQL, MATLAB, Git, GitHub

Machine Learning & Data Science: PyTorch, Scikit-learn, Time Series Analysis, YOLOv8, GenAI, NLTK VADER

Data & Analytical Tools: PowerBI, SQL, Pandas, NumPy, Excel (Pivot Tables, VLOOKUP), MATLAB/Simulink

Business & Communication Tools: MS Office, Agile Project Planning, Stakeholder Communication, Workflow Software

Language: English(7.5 IELTS), German(B1 level), Spanish (A1 level) Telugu (native), Hindi (fluent)

Certifications and Additional Experience

Microsoft PowerBI certification

Nov 2025

- Training in data modelling, KPI dashboards, and interactive visual reports.

Werkraum – Active Project member

Oct 2023 – Present

- Contributed startup ideas and gained hands-on exposure to entrepreneurship through team-based innovation challenges.